# SDK Quick Start Guide
# Configure Beacon Parameters in OTP

## Revision History

| Date | Version | Description |
|---|---|---|
| July 16, 2019 | 0.40 | Initial version created. |
| June 13, 2020 | 0.41 | Minor update to the first two bullets. |
| April 14, 2021 | 0.42 | Updated format, no content change. |

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The ATM2/3 chipset ROM contains a default Bluetooth beacon application with pre-configured parameters. Users may execute the beacon application by ROM and OTP. Currently the beacon application will auto execute upon power up if no flash is present. The beacon parameters can be customized by writing into an on-chip OTP with 4 KB space. This document describes how to write beacon parameters into the OTP.

## Customize Beacon Parameters

- Download the latest SDK: atmosic_sdk_2020xxxx.tar.xz.
- Beacon parameters are configurable under the following path after the SDK is unzipped: \atmosic_sdk\platform\atm2\ATM2xE3c\examples\OTP_beacon\tag_data
- Additional information available in \atmosic_sdk\platform\atm2\ATM2xE3c\examples\OTP_beacon\README
- The beacon parameters are stored in a tag data source (TDS) format in the subfolders.
- Inside the TDS files, every parameter has comments (with “#”) to explain how it is used.
- Users may modify the parameters according to their beacon requirement.

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## Beacon Profile

There is a beacon profile collection, which is convenient for the users to customize their own beacon feature.

- The collection named by reference_beacon.mk is under the path:
  \atmosic_sdk\platform\atm2\ATM2xE3c\examples\OTP_beacon\
- The file refrence_beacon.mk contains several profiles:
  - reference_beacon_power_profile: for measuring the device's low power performance.
  - reference_beacon_extended_adv: extended advertising beacon
  - reference_beacon_2M_secondary: extended beacon via 2 M PHY
  - reference_beacon_LR_coded: extended beacon via long range 125 k PHY
  - reference_beacon_iBeacon: scannable iBeacon
  - reference_beacon_WURX: supports wakeup receiver

## How to set beacon parameter by OTP

The following steps describe how to set up beacon parameters in the OTP.

1. Set up the SDK environment, according to the following two readme file:
   \atmosic_sdk\readme
   \atmosic_sdk\platform\atm2\ATM2xE3c\examples\readme
2. Go to /atmosic_sdk/platform/atm2/ATM2xE3c/examples/OTP_beacon directory

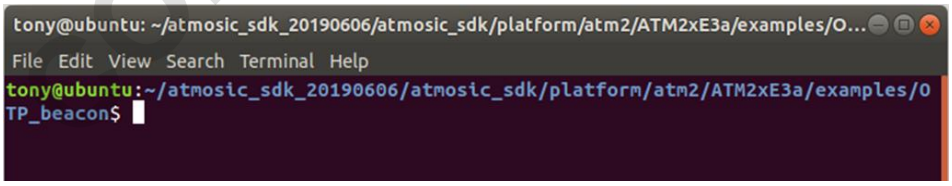



Figure 1 - OTP_beacon Directory

3. Make sure that the Atmosic EVK board is connected to the laptop
4. Read back the existing OTP content of the Atmosic EVK using the following command:
   $ make pull_otp_nvds
   The result is as below:

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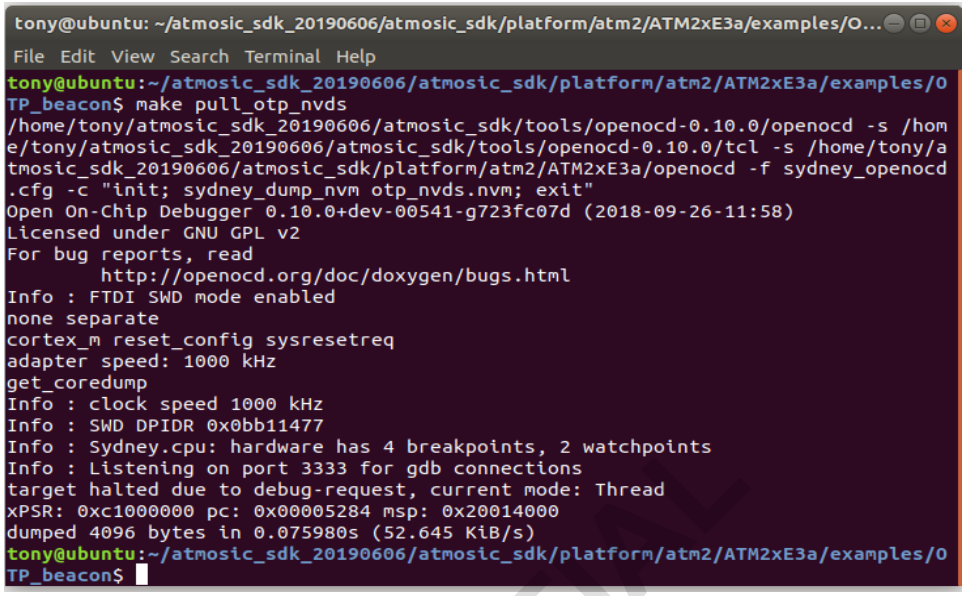



Figure 2 - Existing OTP Content of the EVK

At the same time, otp_nvds.nvm file is generated at \atmosic_sdk\platform\atm2\ATM2xE3c\examples\OTP_beacon

5. Based on otp_nvds.nvm, users may rebuild a new otp file. Users need to make modifications in the .tds and reference_beacon.mk files. At the same time, users also may check OTP content using this command:
   $ make show_pretty_otp_nvds

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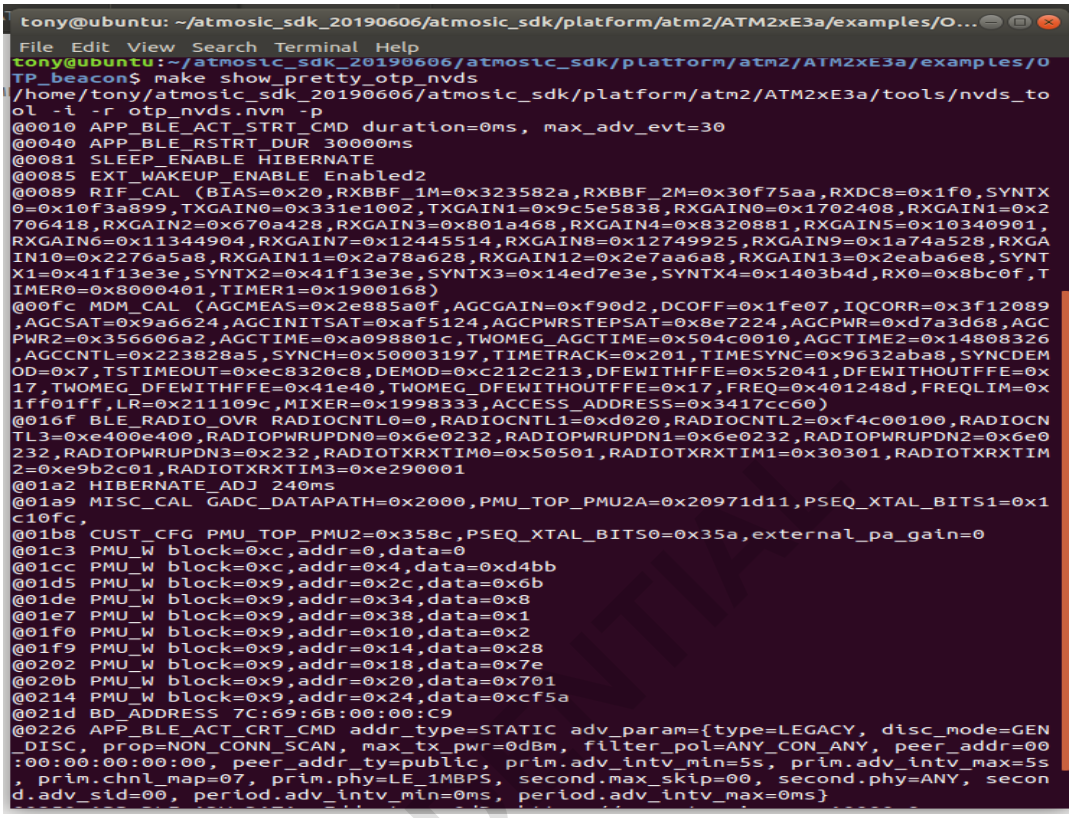



Figure 3 - NVDS OTP Content

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6. Rebuild a new otp_nvds.nvm using the following command:
   $ make REF_BCN:=power_profile rebuild_otp_nvds

```
tony@ubuntu:~/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/examples/O
TP_beacon$ make REF_BCN:=power_profile rebuild_otp_nvds
/home/tony/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/tools/nvds_to
ol -b -i -r otp_nvds.nvm -t 01 -d tag_data/01-BD_ADDRESS/beacon_201.bin -t 06 -d
 tag_data/06-APP_BLE_ACT_CRT_CMD/legacy_nc_s_5s.bin -t 0b -d tag_data/0b-APP_BLE
_ADV_DATA/beacon_201.bin -t 0c -d tag_data/0c-APP_BLE_SCAN_RESP_DATA/beacon_201.
bin -t 05 -d tag_data/05-APP_BLE_ACT_STRT_CMD/30_adv.bin -t 09 -d tag_data/09-AP
P_BLE_RSTRT_DUR/30s.bin -t 11 -d tag_data/11-SLEEP_ENABLE/hib.bin -t 12 -d tag_d
ata/12-EXT_WAKEUP_ENABLE/enable2.bin -t b9 -d tag_data/b9-HIBERNATE_ADJ/rom_bcn.
bin > otp_nvds_new.nvm
mv -f otp_nvds.nvm otp_nvds_df4ecb0b1b0c025bf6496db0ac5efa8f.nvm
tony@ubuntu:~/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/examples/O
tony@ubuntu:~/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/examples/O
TP_beacon$
```

Figure 4 - Rebuild New OTP NVDS Content

7. Burn the parameters into OTP using the following command:
   $ make push_otp_nvds

```
tony@ubuntu:~/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/examples/O
TP_beacon$ make push_otp_nvds
/home/tony/atmosic_sdk_20190606/atmosic_sdk/tools/openocd-0.10.0/openocd -s /hom
e/tony/atmosic_sdk_20190606/atmosic_sdk/tools/openocd-0.10.0/tcl -s /home/tony/a
tmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/openocd -f sydney_openocd
.cfg -c "init; sydney_burn_nvm otp_nvds.nvm; sydney_verify_nvm otp_nvds.nvm; exi
t"
Open On-Chip Debugger 0.10.0+dev-00541-g723fc07d (2018-09-26-11:58)
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Info : FTDI SWD mode enabled
none separate
cortex_m reset_config sysresetreq
adapter speed: 1000 kHz
get_coredump
Info : clock speed 1000 kHz
Info : SWD DPIDR 0x0bb11477
Info : Sydney.cpu: hardware has 4 breakpoints, 2 watchpoints
Info : Listening on port 3333 for gdb connections
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005284 msp: 0x20014000
4096 bytes written at address 0x20000000
downloaded 4096 bytes in 0.064348s (62.162 KiB/s)
target halted due to debug-request, current mode: Thread
xPSR: 0xc1000000 pc: 0x00005284 msp: 0x20014000
4096 bytes written at address 0x20000000
downloaded 4096 bytes in 0.061078s (65.490 KiB/s)
tony@ubuntu:~/atmosic_sdk_20190606/atmosic_sdk/platform/atm2/ATM2xE3a/examples/O
TP_beacon$
```

Figure 5 - Burn Parameters into OTP

## Summary

There are mainly four steps to handle the beacon parameters OTP update

1. Use command:
   make pull_otp_nvds
   to get the original OTP content (otp_nvds_nvm).
2. Change the beacon parameters in .tds, and reference_beacon.mk, according to the requirement
3. Rebuild the OTP content:
   make REF_BCN:=power_profile rebuild_otp_nvds
4. Burn content into OTP:
   make push_otp_nvds

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# Background Information

**Datasheet/Reference Manual**

- IC specification is provided in the datasheet and more detailed information is given in the reference manual.

**Evaluation Kit (EVK)**

- The hardware evaluation kit consists of two boards:
  - An interface board to connect the Atmosic Development board to a laptop. Details are given in the Interface Board User Guide in the ATM3_2X/Docs folder.
  - A development board with the Atmosic IC. Details on this board are given in the EVK User Guide in the ATM3_2X folder.

**Software Development Kit (SDK)**

- To get started with software development kit (SDK), refer to the SDK quick start guide in the ATM3_2X/Docs folder.

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